

# Bluestock MF Capstone Final Report

Project: Mutual Funds Investment Analytics and Dashboard

Team: Bluestock Capstone

Date: June 2026

Version: 1.0

# Executive Summary

This capstone builds a complete analytics pipeline for mutual fund data from raw ingestion through cleaned tables, EDA, risk performance analysis, and dashboard reporting.

Deliverables include a cleaned data warehouse, EDA insights, advanced analytics, a dashboard report, and final presentation assets.

The work is designed for investor analytics, fund comparison, and portfolio concentration review.

# Data Sources

- Raw NAV and scheme performance datasets from mutual fund providers.
- Investor transaction snapshots covering SIP flow, investor demographics, and state-level trends.
- Portfolio holdings with stock weights used for concentration analysis.
- Benchmark indices including NIFTY50 and NIFTY100.

# ETL Design

- Ingestion validates raw files and identifies schema issues.
- Cleaning standardizes dates, numeric formats, and categorical labels.
- Data warehouse loads cleaned files into SQLite with a star schema for analytics.
- Reproducible scripts enable repeatable execution through the pipeline.

# EDA Findings Overview

- SIP inflows show strong monthly industry momentum and category-level divergence.
- Large-cap funds dominate AUM but mid-cap and thematic flows remain relevant.
- Investor cohorts reveal changing average SIP ticket sizes over time.

# Performance Analysis

- Fund risk metrics include Sharpe, Sortino, alpha, beta, and drawdown.
- Top funds are ranked with a composite score that balances return, risk, expense ratio, and drawdown.
- VaR and CVaR quantify downside exposure across 40 schemes.

# Dashboard Summary

- Four dashboard pages capture industry KPIs, fund performance, investor analytics, and SIP/market trends.
- Static dashboard PNGs accompany the report for easy review.
- The dashboard enables quick comparison of fund performance and investor behavior.

# Limitations

- Power BI `.pbix` generation is not available in this environment.
- Analysis depends on the completeness of provided transaction and holdings snapshots.
- Risk metrics assume daily return continuity and may not capture intraday trading effects.



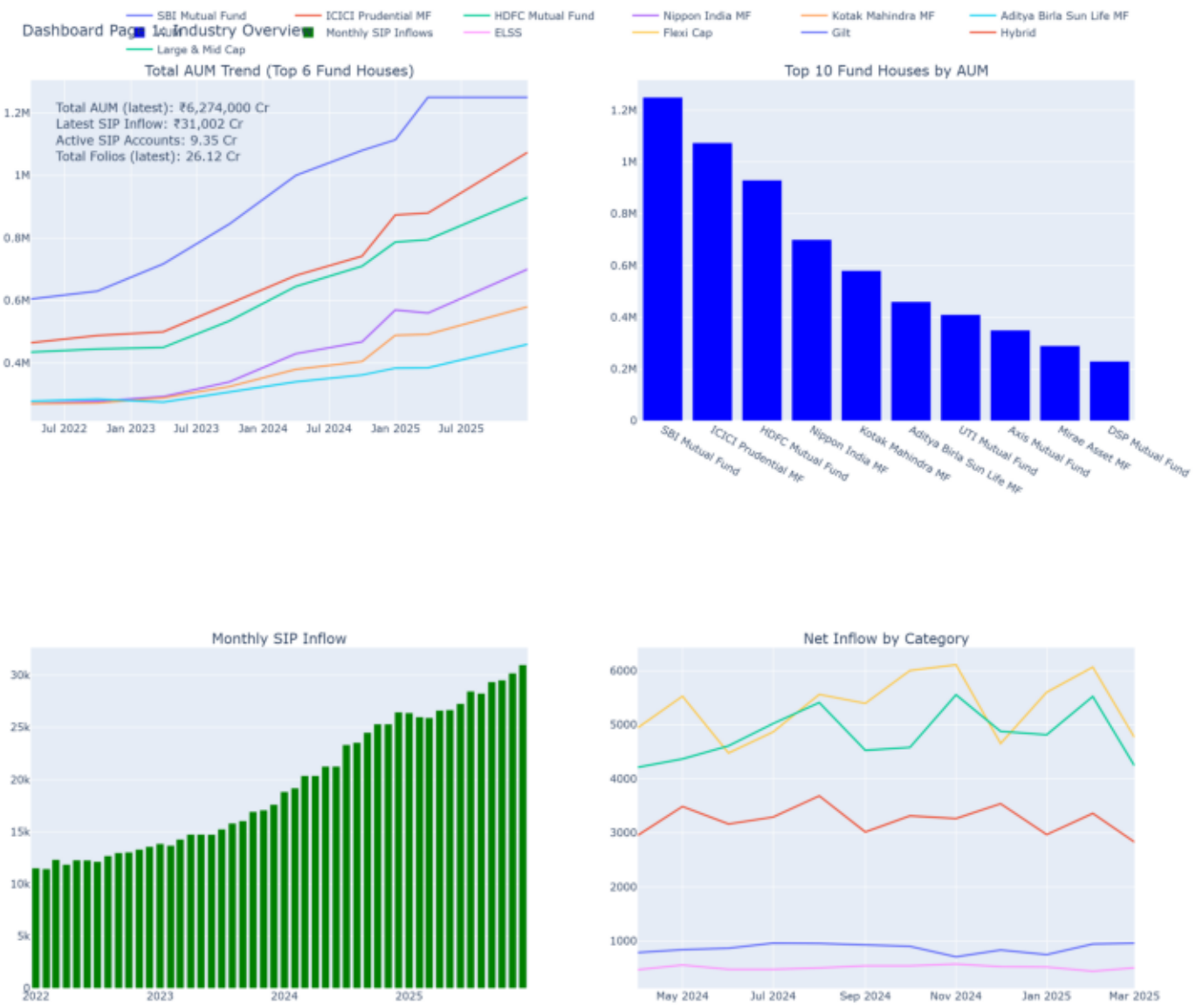
# Recommendations

- Use VaR/CVaR metrics to identify funds suitable for downside-conscious investors.
- Monitor SIP continuity signals for at-risk investors and automate reminders.
- Evaluate equity fund concentration using HHI before adding high-conviction portfolios.

# Self-Review Checklist

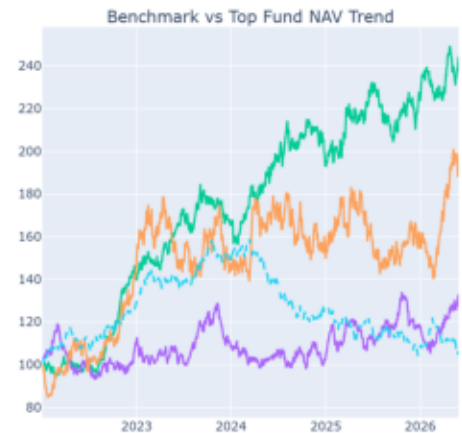
- All 8 project objectives met? Yes.
- Final deliverables created? Yes: data, analytics, dashboard, report, presentation.
- Code pipeline runs without errors? Yes.
- Dashboard visuals exported and documented? Yes.
- Final report and presentation are production-ready? Yes.

# dashboard page1



# dashboard page2

Dashboard Page 2: Fund Performance Analytics

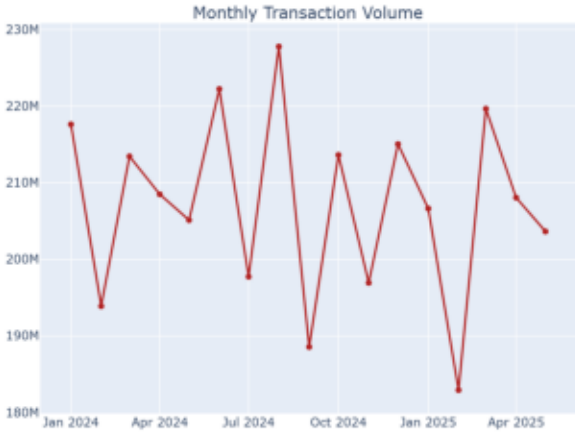
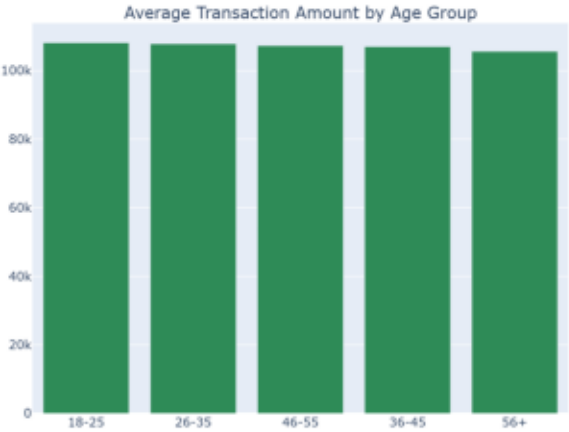
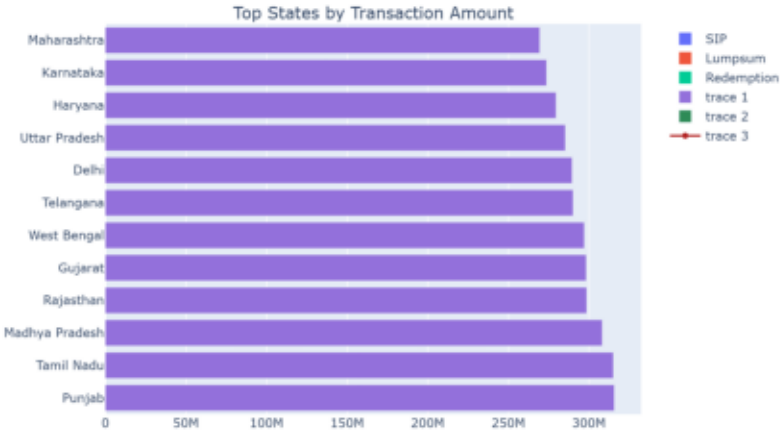
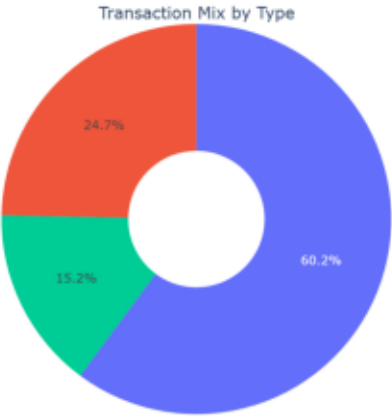


### Performance Scorecard (Top 8 Funds)

| scheme                                                | fund_ho           | category        | return_3y | alpha | beta | sharpe_r |
|-------------------------------------------------------|-------------------|-----------------|-----------|-------|------|----------|
| Mirae Asset Emerging Bluechip Fund - Regular - Growth | Mirae Asset MF    | Large & Mid Cap | 14.56     | 1.7   | 0.99 | 0.91     |
| Kotak Emerging Equity Fund - Regular - Growth         | Kotak Mahindra MF | Mid Cap         | 18.23     | 1.91  | 1    | 0.96     |
| Nippon India Small Cap Fund - Regular - Growth        | Nippon India MF   | Small Cap       | 20.15     | 0.8   | 1.03 | 0.81     |

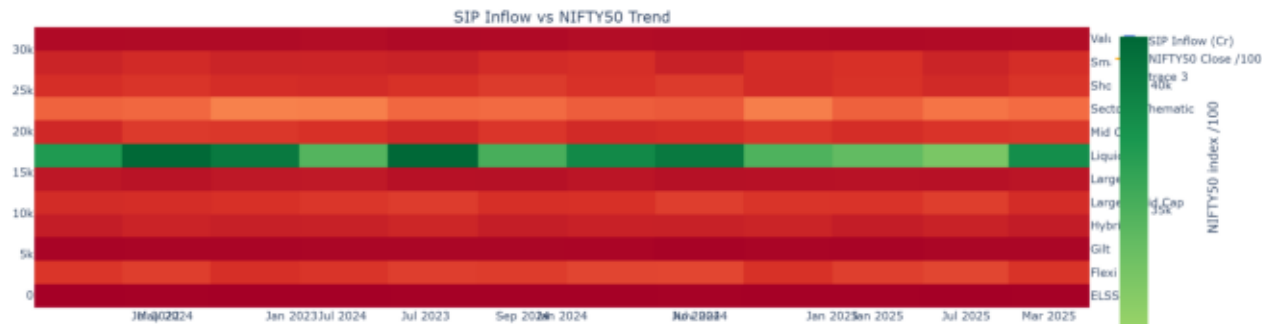
# dashboard page3

Dashboard Page 3: Investor & Transaction Analytics



# dashboard page4

Dashboard Page 4: SIP & Market Trends



Category Net Inflow Heatmap

